

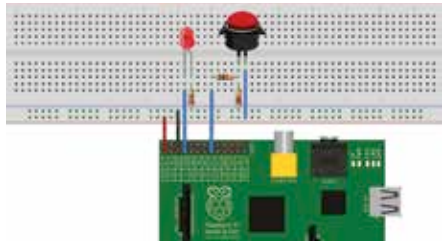
The Raspberry Pi has pins called ‘GPIO [General Purpose Input Output] pins’ that allow switching ON and OFF. To use the GPIO pins, you need to install the GPIO Python library that allows you to switch the voltage level on a particular GPIO pin. This change in voltage level can switch appliances ON or OFF by using relays.

Software needed:

To control the voltage of the GPIO pin, you’ll need to install the GPIO library that allows voltage control using the Python programming language.

To install the GPIO Python library, open the terminal and type the following commands:

```
• sudo apt-get install Python-dev
• sudo apt-get
install Python-rpi.
gpio
```



Hardware required:

- ◆ Raspberry Pi
- ◆ Breadboard
- ◆ Wires
- ◆ LED
- ◆ 1K resistor – Three Circuit for blinking LED in number
- ◆ Switch

Connect the circuit as shown in the given figure. If you’re new to electronics, we advise taking the help of someone with some knowledge of electronic circuits.

For the first part, we’ll just switch the LED ON and OFF alternatively.

We’ll write the Python code for the same as follows:

```
• import RPi.GPIO as GPIO
• import time

• GPIO.setmode(GPIO.BCM)
• GPIO.setup(4, GPIO.OUT)
• while True:
```